

Qilin Huang

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Seeking Summer 2027 Internships

EDUCATION

Carnegie Mellon University

Master of Science in Computer Vision

Current Coursework: Advanced Computer Vision; Visual Learning and Recognition; Introduction to Robot Learning.

Pittsburgh, PA

Expected Dec. 2027

Southern University of Science and Technology

Bachelor of Engineering in Computer Science and Technology

GPA: 3.89/4.00 | **Rank:** 13/167

Selected Coursework: Natural Language Processing; Intelligent Robotics.

Shenzhen, China

Jun. 2026

TECHNICAL SKILLS

Programming: Python, Java, C/C++, SQL.

Libraries & Tools: PyTorch, NumPy, scikit-learn, Linux, SLURM.

RESEARCH & PROJECTS

UniPixie: Controllable 3D Physics

Research Project | GRASP Laboratory, University of Pennsylvania

CVPR 2026 Highlight

Jul. 2025 - Mar. 2026

Philadelphia, PA / Remote

- Built a **conditional flow-matching model** to infer spatially varying material properties from visual features, with scalar conditioning for **continuous stiffness control**.
- Designed comparative experiments and ablation studies; UniPixie achieved **55.6% lower** MSE in log Young's modulus than retrained PIXIE on PixieMultiVerse.
- Developed unified interfaces connecting model predictions to **three existing simulation backends**: Material Point Method, linear blend skinning, and spring-mass systems.

Neural Module Network for 3D Visual Question Answering

Independent Research Project | Southern University of Science and Technology

- Proposed and implemented a 3D VQA prototype using a frozen LLM to translate natural-language questions into executable programs.

- Integrated task-specific 3D modules for step-by-step reasoning and evaluated the approach on ScanQA.

Feb. - Jun. 2025

Shenzhen, China

HCNQA: 3D Visual Question Answering

Research Project | Southern University of Science and Technology

- Reproduced baseline experiments on ScanQA for comparative evaluation and authored the VQA related-work section.

Jun. - Sep. 2024

Shenzhen, China

CPU/GPU Matrix Operations and Performance Evaluation

C++ Programming Course Project | Southern University of Science and Technology

- Implemented CUDA elementwise matrix operations and integrated OpenBLAS/cuBLAS for matrix multiplication; compared CPU/GPU runtimes across matrix sizes.

Jun. 2024

Shenzhen, China

PUBLICATIONS

UniPixie: Unified and Probabilistic 3D Physics Learning via Flow Matching

Qilin Huang et al. **CVPR 2026, Highlight.** *Co-first author.*

HCNQA: Enhancing 3D VQA with Hierarchical Concentration Narrowing Supervision

Shengli Zhou, Jianuo Zhu, **Qilin Huang**, et al. **ICANN 2025.**